

EBR-II SHRT-17 XX09 Protected Loss-of-Flow Transient Study

MOOSE/SubChannel – Part B

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Independent portfolio study

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Study aim. Extend the Part A steady-state XX09 study to the 900 s SHRT-17 protected loss-of-flow transient, reproduce the public MOOSE/SubChannel calculation with the current upgraded Cheng–Todreas closure, examine the imposed power and flow histories, and compare the predicted TTC-31 temperature response with published experimental measurements digitized from the official MOOSE validation figure.

1 Benchmark and transient scope

SHRT-17 is an EBR-II shutdown-heat-removal experiment initiated from full-power operation as a protected loss-of-flow event. The benchmark specification documents the initial conditions, pump coastdown, reactor-power history, instrumentation, and quantities used for code comparison [1]. Part A of this portfolio established the pre-transient XX09 subassembly operating point. Part B keeps the same local subchannel representation and follows its thermal response through the subsequent 900 s transient.

This is not a full-plant EBR-II calculation. In the public standalone SubChannel validation problem, the XX09 mass-flow and power histories are imposed as functions of time [2]. The calculation therefore evaluates the bundle-scale thermal response to benchmark forcing; it does not independently predict the system-level pump coastdown or natural-circulation mass flow.

Table 1: Principal transient-model settings used in the present calculation.

Item	Configuration
Application / problem	MOOSE SubChannelApp; TriSubChannel1PhaseProblem
MOOSE package	2026.09.25
Fluid properties	PBSodiumFluidProperties [3]
Axial discretization	50 cells
Friction closure	SCMFrictionUpgradedChengTodreas [4]
Heat-transfer closure	Gnielinski [5]
Mixing closure	Cheng–Todreas, $C_T = 2.6$
Transient interval	$t = -1$ to 900 s; adaptive time stepping; $\Delta t_{\max} = 20$ s
Outlet pressure / inlet temperature	200 kPa / 624.7 K
Initial XX09 power / mass flow	486.2 kW / 2.45 kg/s

2 Numerical setup, provenance, and execution

The public MOOSE SHRT-17 transient input was used as the numerical starting point [2]. Unlike the older software snapshot used in Part A, the MOOSE 2026.09.25 environment accepted the upstream `SCMFrictionUpgradedChengTodreas` object directly, so no friction-closure substitution was required in the final Part B production case. The original fluid, friction, heat-transfer, mixing, geometry, power, and inlet-flow definitions were retained.

For the production run, only the `MultiApps` and `Transfers` blocks used to project the solution onto a separate detailed visualization mesh were removed. This reduced computational overhead without altering the parent `SubChannel` equations or the prescribed transient forcing. The run reached 900 s in 91 adaptive timesteps and completed successfully. One initialization warning reported the bulk Reynolds number outside the upgraded Cheng–Todreas correlation data range at Step 0; no repeated warning was reported during the subsequent transient.

Listing 1: Core transient time-stepping settings retained from the public validation input.

```
[Executioner]
  type = Transient
  start_time = -1.0
  end_time = 900.0
[TimeStepper]
  type = IterationAdaptiveDT
  dt = 0.1
  iteration_window = 5
  optimal_iterations = 6
  growth_factor = 1.1
  cutback_factor = 0.8
  timestep_limiting_function = 'time_step_limiting'
[]
dtmax = 20
[]
```

3 Transient forcing and physical sequence

The imposed reactor-power history falls much faster than the assembly flow immediately after the transient begins. Power decreases from 486.2 kW at $t = 0$ to approximately 96.4 kW at 1 s, while mass flow remains about 2.26 kg/s. During the following pump coastdown, however, the flow becomes very small: interpolation of the prescribed history gives approximately 0.0536 kg/s at 60 s. By then decay power is still approximately 18.6 kW. The normalized histories in Fig. 1 make this changing balance between heat generation and coolant transport explicit.

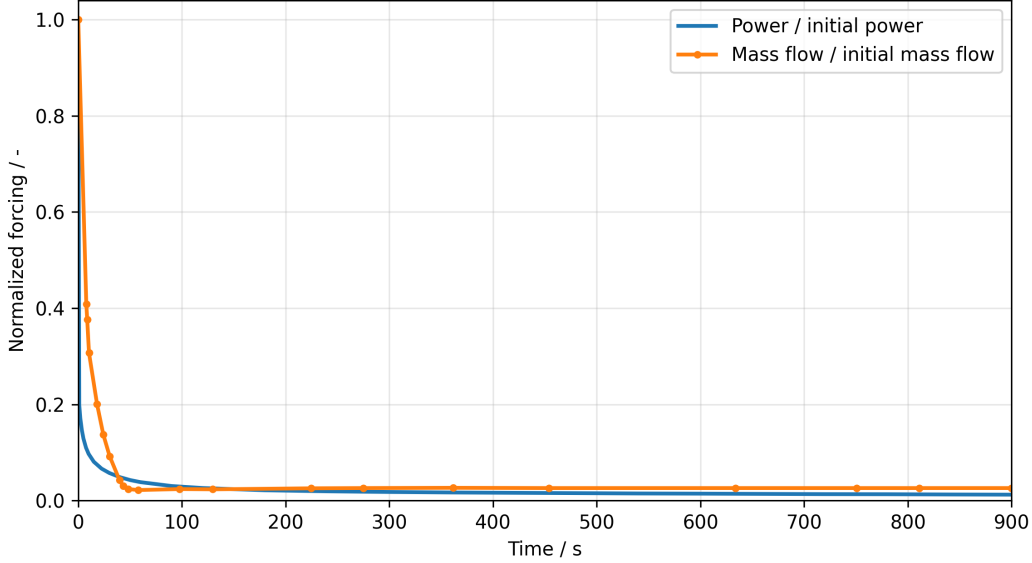


Figure 1: Prescribed SHRT-17 power and XX09 mass-flow histories used in the standalone SubChannel transient. Both quantities are normalized by their initial values.

A compact way to interpret the sequence is the instantaneous power-to-flow ratio, $\dot{Q}/\dot{m}$. It falls from about 198.4 kJ/kg initially to 42.6 kJ/kg at 1 s, consistent with the rapid early coolant cooling. By 60 s, the same ratio has risen to approximately 346.5 kJ/kg because the mass flow has collapsed more strongly than the remaining decay power. This helps explain the subsequent reheating of TTC-31 before the long-term cooling phase.

Table 2: Selected values from the prescribed forcing histories. Mass flow is linearly interpolated from the public history file.

t [s]	Power [kW]	$\dot{m}$ [kg/s]	$\dot{m}/\dot{m}_0$	$\dot{Q}/\dot{m}$ [kJ/kg]
0	486.20	2.4500	1.0000	198.45
1	96.41	2.2638	0.9240	42.59
10	47.19	0.7888	0.3220	59.82
30	27.84	0.2309	0.0942	120.58
60	18.59	0.0536	0.0219	346.54
300	8.66	0.0639	0.0261	135.47
900	5.94	0.0635	0.0259	93.51

4 TTC-31 transient response and experimental comparison

The final pre-transient output sample gives TTC-31 = 813.54 K at $t = -0.051$ s. After the scram, the calculated temperature decreases rapidly to 660.25 K at 3.56 s. Continued flow coastdown then reduces heat removal enough for TTC-31 to reheat, reaching a calculated maximum of 883.09 K at 64.26 s. Thereafter the temperature decreases gradually with the continuing decay-power reduction, ending at 696.57 K at 900 s.

For comparison, the experimental TTC-31 markers were digitized from the official MOOSE SHRT-17 validation figure [2]. They are therefore treated as figure-derived values rather than raw experimental data. The digitized experimental peak is approximately 892.12 K at 74.0 s. Relative to that point, the present MOOSE calculation predicts the peak 9.03 K lower and 9.74 s earlier.

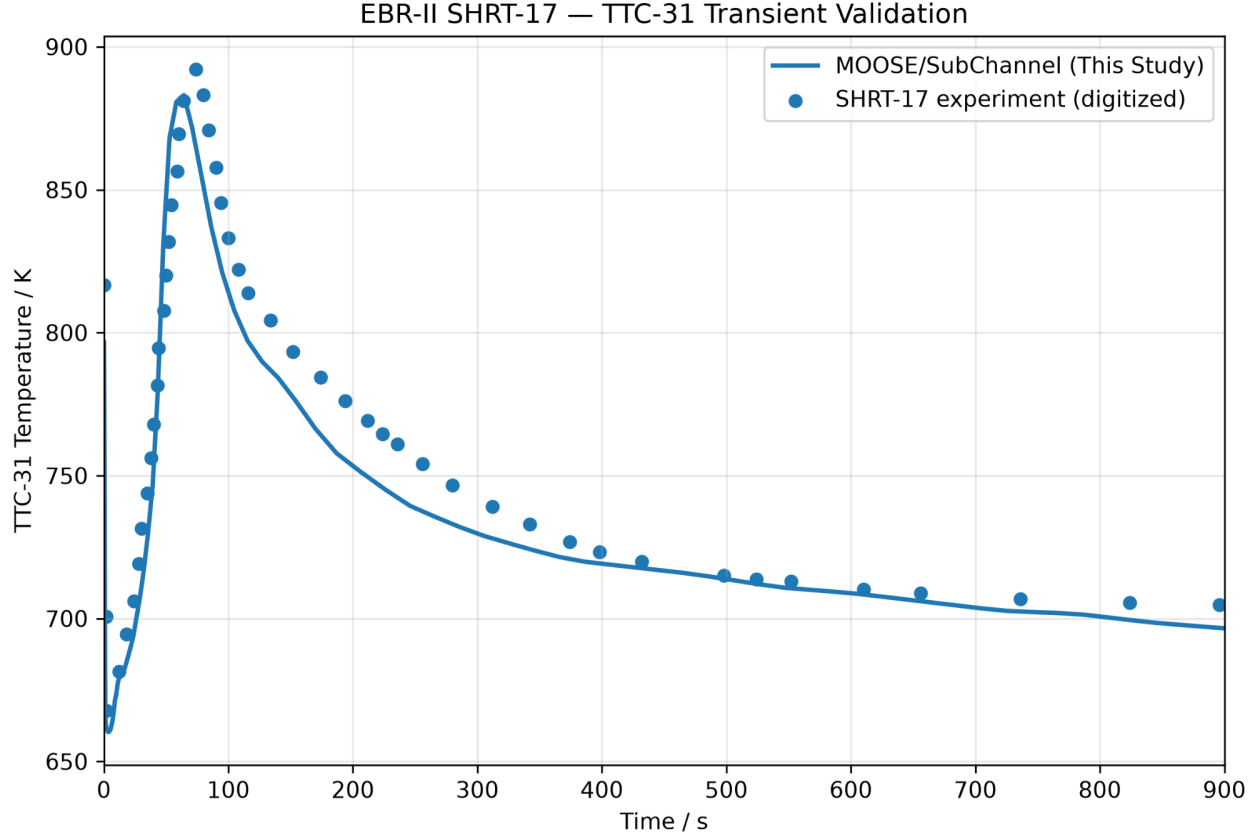


Figure 2: TTC-31 transient temperature from the present MOOSE/SubChannel run and experimental markers digitized from the published MOOSE SHRT-17 validation figure [2].

The simulation was interpolated to the 50 digitized experimental measurement times. For paired values T_i^M and T_i^E , the reported summary errors are

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |T_i^M - T_i^E|, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (T_i^M - T_i^E)^2}. \quad (1)$$

The resulting values are $\text{MAE} = 14.88 \text{ K}$ and $\text{RMSE} = 17.65 \text{ K}$. The calculation reproduces the main qualitative sequence and the long cooling tail, while the predicted transient peak is somewhat lower and earlier than the digitized measurement peak.

Table 3: Headline TTC-31 transient comparison.

Quantity	Result
Pre-transient TTC-31	813.54 K
Post-scam minimum	660.25 K at 3.56 s
MOOSE peak	883.09 K at 64.26 s
Digitized experimental peak	892.12 K at 74.0 s
Peak-temperature difference	−9.03 K
Peak-time difference	−9.74 s
MAE / RMSE	14.88 K / 17.65 K
Final TTC-31 at 900 s	696.57 K

5 Interpretation, limitations, and conclusions

Part B reproduces the public SHRT-17 XX09 transient with the current upgraded Cheng–Todreas friction closure and provides a transparent code-to-data check of TTC-31. The calculated temperature exhibits the expected three-stage behavior: a rapid post-scam decrease as power collapses, reheating as the imposed flow coastdown becomes dominant, and a long decay-heat cooling tail. The predicted peak is about 1.0% below the digitized experimental peak and occurs about 9.7 s earlier; the full set of digitized points gives $\text{MAE} = 14.88 \text{ K}$ and $\text{RMSE} = 17.65 \text{ K}$.

The result should be interpreted within the model scope. The standalone XX09 calculation uses prescribed power and mass-flow histories, so it does not constitute an independent system-level prediction of pump coastdown or natural

circulation. The comparison data were digitized from a published figure, not obtained from the original measurement file, and therefore include small digitization uncertainty. The calculation also represents one instrumented subassembly rather than the full EBR-II primary system. These limitations make the study suitable as a reproducible portfolio validation exercise, not as a licensing-grade safety analysis.

Provenance statement. Argonne provides the SHRT-17 benchmark specification [1]; the MOOSE project provides the public XX09 transient validation setup and published comparison figure [2]. For this study, the transient was executed in MOOSE 2026.09.25, the visualization-only MultiApp/transfer blocks were removed from the production input, transient quantities were extracted and interpreted, the published TTC-31 measurements were digitized, error metrics were calculated, and the figures and technical note were produced independently.

References

- [1] T. Sumner and T. Y. C. Wei. *Benchmark Specifications and Data Requirements for EBR-II Shutdown Heat Removal Tests SHRT-17 and SHRT-45R*. Tech. rep. ANL-ARC-226 Rev.1. Argonne National Laboratory, May 2012. URL: <https://publications.anl.gov/anlpubs/2012/06/73647.pdf>.
- [2] MOOSE Framework. *EBR-II, SHRT-17 Validation*. Idaho National Laboratory MOOSE documentation. Accessed 2026-10-02. URL: <https://mooseframework.inl.gov/modules/subchannel/v%26v/EBR-II.html>.
- [3] MOOSE Framework. *PBSodiumFluidProperties*. Idaho National Laboratory MOOSE documentation. Accessed 2026-10-02. URL: <https://mooseframework.inl.gov/source/fluidproperties/PBSodiumFluidProperties.html>.
- [4] MOOSE Framework. *SCMFrictionUpgradedChengTodreas*. Idaho National Laboratory MOOSE documentation. Accessed 2026-10-02. URL: <https://mooseframework.inl.gov/moose/source/scmclosures/SCMFrictionUpgradedChengTodreas.html>.
- [5] V. Gnielinski. “Neue Gleichungen für den Wärme- und den Stoffübergang in turbulent durchströmten Rohren und Kanälen”. In: *Forschung im Ingenieurwesen A* 41.1 (1975), pp. 8–16.